CBCS SCHEME

USN | | | | | | | | | | |

BBOC407

Fourth Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Biology for Engineers (CSE)

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom s level , C: Course outcomes.

		Module 1	M	L	C
Q.1	a.	Explain the structure and function of power house of cell and Endoplasmic reticulum with neat diagram.	10	L2	CO1
	b.	What are stem cells? Explain the properties classification and application of stem cells.	10	L1	CO1
		OR			
Q.2	a.	Explain the structures properties and function of nucleic acid focusing on DNA.	10	L2	CO1
	b.	Define Vitamins. Explain the properties, function, source and deficiency of vitamins.	10	L1	CO1
		Module 2			
Q.3	a.	Illustrate the steps involved in biodiesel production. Add a note principle and limitation of biodiesel.	10	L2	CO2
	b.	Develop the protocol for PLA polymer synthesis. Add a note on engineering application of PLA.	10	L3	CO2
		OR			
Q.4	a.	Define Biosensor. Outline the principle, working and application of enzyme in glucose biosensor.	10	L2	CO2
	b.	Construct the procedure for the production of RNA vaccines against Covid-19. Add a note on how RNA vaccines different from DNA vaccines.	10	L3	CO2
		Module 3			
Q.5	a.	Compare and contract brain as CPU system and eye as a camera.	10	L2	CO3
	b.	Explain the mechanism of filtration in Human Kidney.	10	L2	CO3
		OR			
Q.6	a.	Write in detail Heart Lung Machine.	10	L2	CO3
	b.	Explain how lung act as purification system. Add a note on principle and working of spirometry as a diagnostic tool for assessing lung function.	10	L2	CO3

		Module 4			
Q.7	**a.**	Apply the concept of bioecholocation in the field of Navigation and detection. Write the principle, working and instrumentation and application of the technique.	**10**	**L3**	**CO3**
	b.	HBOCs and PFCs act as human blood substitutes. Explain.	**10**	**L4**	**CO3**
		OR			
Q.8	**a.**	Identify and explain the process, application of technique involved in conversion of light energy into electric energy.	**10**	**L3**	**CO3**
	b.	Velcro and friction less swimsuits are the nature bioinspired material. Explain the principle and engineering application of the technology.	**10**	**L4**	**CO3**
		Module 5			
Q.9	**a.**	Apply the process of biomining via microbial surface adsorption for the removal of heavy metals.	**10**	**L3**	**CO4**
	b.	Analyze the principle, working and instrumentation of e-tongue, highlighting its application in food and beverage industries.	**10**	**L4**	**CO4**
		OR			
Q.10	**a.**	Develop the steps for 3D printing of skin. Highlight on materials used and application of 3D skin.	**10**	**L3**	**CO4**
	b.	Bio imaging and artificial intelligence technique plays important role in disease diagnosis. Explain the concept and add a note on its limitation.	**10**	**L4**	**CO4**

* * * * *

VTU - 28-11-2024 01:18:26pm - VC - VC - VC

CBCS SCHEME

4/19

BBOC407

USN

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Biology for Engineers (CSE)

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is stem cell? Explain its types and list its applications.	7	L2	CO1
	b.	Explain in detail the properties and functions of nucleic acids.	6	L2	CO1
	c.	Explain the importance of special biomolecules.	7	L2	CO1
		OR			
Q.2	a.	What is a biomolecule? Explain the classifications of biomolecule.	7	L2	CO1
	b.	Explain the properties and functions of carbohydrates.	6	L2	CO1
	c.	Describe the structure and functions of a cell with a neat diagram.	7	L3	CO1
		Module – 2			
Q.3	a.	What is the role of lipids? Outline the process of obtaining biodiesel from lipids.	7	L3	CO2
	b.	Differentiate between PHA and PLA as a bioplastic materials.	6	L4	CO1
	c.	Explain the role of DNA vaccine for rabies and RNA vaccine for COVID-19.	7	L2	CO1
		OR			
Q.4	a.	What are the key properties, advantages and limitations of cellulose based water filters.	7	L3	CO2
	b.	How can DNA finger printing be applied to evaluate its effectiveness and reliability in forensic applications.	6	L4	CO1
	c.	Describe the use of meat analogue and plant protein as food.	7	L2	CO2
		Module – 3			
Q.5	a.	Deliberate the functioning of brain as CPU system.	7	L3	CO2
	b.	Write a short note on spirometry and ventilator.	6	L2	CO2
	c.	Explain heart as pump system.	7	L3	CO2

		OR			
Q.6	**a.**	Explain eye as a camera system.	**7**	**L3**	**CO2**
	b.	Write a short note on cardiac pacemaker.	**6**	**L2**	**CO2**
	c.	Explain kidney as purification system.	**7**	**L3**	**CO2**
		Module – 4			
Q.7	**a.**	Describe the materials used and engineering applications of Velcro technology.	**7**	**L3**	**CO3**
	b.	Compare the process of photosynthesis to the functioning of photovoltaic cells.	**6**	**L4**	**CO3**
	c.	Explain the HBOCs and PFCs as human blood substituents.	**7**	**L3**	**CO3**
		OR			
Q.8	**a.**	Explain the terms lotus leaf effect and bird flying.	**7**	**L3**	**CO3**
	b.	Compare biological echolocation and technological echolocation highlighting their applications in navigation and detection.	**6**	**L4**	**CO3**
	c.	Explain the terms shark skin, swim suits and bullet train using biological concepts.	**7**	**L3**	**CO3**
		Module – 5			
Q.9	**a.**	Compare the functioning of electrical tongue and human tongue.	**7**	**L4**	**CO4**
	b.	Explain muscle cells as scaffold for tissue growth.	**6**	**L2**	**CO4**
	c.	Explain bioremediation and biomining via microbial surface adsorption.	**7**	**L2**	**CO4**
		OR			
Q.10	**a.**	Illustrate the basic steps of bioprinting process and list the various types of bioprinting techniques.	**7**	**L4**	**CO4**
	b.	Write a short note on: i) Importance of DNA origami ii) Self healing bioconcrete.	**6**	**L2**	**CO4**
	c.	Discuss the applications of artificial intelligence in the diagnosis of disease.	**7**	**L2**	**CO4**

CMRIT LIBRARY
BANGALORE - 560 037

* * * * *

CBCS SCHEME

USN | | | | | | | | | | |

BBOC407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Biology for Engineers (CSE)

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Discuss the various components of Eukaryotic cells.	10	L3	CO1
	b.	Identify the applications of stem cells.	5	L2	CO1
	c.	Explain the functions of vitamins.	5	L2	CO1
		OR			
Q.2	a.	Compare Prokaryotic and Eukaryotic cells.	10	L3	CO1
	b.	Explain the properties of Carbohydrates.	5	L2	CO1
	c.	Explain the functions of Lipids.	5	L2	CO1
		Module – 2			
Q.3	a.	Highlighting the properties of cellulose, justify cellulose as an effective water filter.	10	L3	CO1
	b.	Explain the working and development of DNA vaccines by taking suitable example.	10	L2	CO1
		OR			
Q.4	a.	What are Bioplastics? Justify the use of PHA as Bioplastic mentioning its properties and applications.	10	L3	CO1
	b.	Discuss the following : (i) Meat analogs of protein. (ii) Lipids as cleaning agents.	10	L2	CO1
		Module – 3			
Q.5	a.	What is Electro Encephalogram (EEG)? Discuss the types of Brain activity detected with EEG. Write any three applications.	10	L3	CO2
	b.	What are Pace Makers? Explain basic design and construction of Pace Makers.	10	L2	CO2
		OR			
Q.6	a.	Justify Lungs as purification system.	10	L3	CO2
	b.	Explain architecture of Rod and Core cells with suitable diagram.	10	L2	CO2
		Module – 4			
Q.7	a.	What is ultrasonography? Explain the uses and working principle.	10	L2	CO3
	b.	What is lotus leaf effect? Explain the mechanism and applications of super Hydrophobic effect.	10	L2	CO3
		OR			
Q.8	a.	The structure and design of Kingfisher beak lead to the design of Bullet trains. Explain.	10	L2	CO3
	b.	Explain the working and applications of Bionic Leaf Technology.	10	L2	CO3

		Module – 5			
Q.9	**a.**	Explain the use of Electrical tongue in food science.	**10**	**L2**	**CO4**
	b.	Explain the advantages and limitations of Artificial Intelligence for disease diagnosis.	**10**	**L2**	**CO4**
		OR			
Q.10	**a.**	Explain Bioengineering solutions for muscular dystrophy and Osteroporosis.	**10**	**L2**	**CO4**
	b.	Explain most commonly used Bioprinting Techniques.	**10**	**L2**	**CO4**

* * * * *